

Lab 11: Descriptive Statistics, Correlation and Regression

Level: Advanced Mapping: LO6 | K1 | K2 | A1 | A3 | A4 | A6 Concepts: descriptive statistics, correlation, lm, residuals, R-squared

Scenario

A health analytics team studies height, weight and a numeric outcome.

Goal

A validated summary, correlation result, fitted model and diagnostic evidence.

Files in this folder

- lab-11.R - completed guided R script
- verify.R - repeatable acceptance check
- lab-workbook.xlsx - brief, input data, controls and evidence record
- input-data.csv - portable fallback if readxl is unavailable

Detailed step-by-step

1. Import the health dataset from the workbook.
2. Describe valid sample size, centre and spread.
3. Plot height versus weight and calculate Pearson correlation.
4. Fit a linear model and interpret slope and R-squared.
5. Inspect residuals and record one limitation.

Run and verify

1. Set the RStudio working directory to this lab folder.
2. Open lab-11.R and read the header, inputs and validation rules.
3. Run the complete script with `source("lab-11.R")`.
4. Confirm the Console ends with LAB VERIFIED: outputs created and checks passed.
5. Run `source("verify.R")` and retain the verification message.
6. Open `analysis_output.csv` and compare the evidence with the workbook Control sheet.
7. Complete the workbook Evidence sheet and save it without changing the filename.

Acceptance criteria

- [] Correlation interpretation includes direction and strength.
- [] Model formula and prediction target are explicit.
- [] Diagnostic evidence is included before a conclusion.

Troubleshooting

- there is no package called 'readxl': the script automatically uses input-data.csv; install readxl later if Excel import is required.
- cannot open file: confirm RStudio's working directory is this exact lab folder.
- object not found: run the script from the top so imports and validations occur before analysis.
- Verification fails: inspect the first error, compare column names with Input_Data, and rerun from a clean session.

Evidence to submit

Completed Excel Evidence sheet, R script, generated output files and the final Console verification message.